

Generation Interconnection System Impact Study Report

For Queue Project Q14-0355

Altavista 138 kV

August 31, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0355. The Interconnection Customer proposes a 1075.8 MW natural gas combined cycle facility interconnecting at the Altavista 138 kV substation in the Delaware area. The request was received on July 22, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 152 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$147,096,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of VAN SANT STA / ALTAVISTA SUBSTATION 138.0 kV, the ALTAVISTA / BRISTOL 138.0 kV L loads to 384.7% of its 269 MVA rating. Project DFAX: 100.0%.

ALTAVISTA / VAN SANT STA 138.0 kV: 371.0% loading (279 MVA) following the outage of the ALTAVISTA / BRISTOL SUBSTATION 138.0 kV; DFAX = 100.0%.

SHOALS / BRISTOL 138.0 kV: 305.4% loading (164 MVA) following the outage of the VAN SANT STA / ALTAVISTA 138.0 kV L; DFAX = 49.3%.

For the loss of BRISTOL / ALTAVISTA 138.0 kV, the AVOCA / VAN SANT STA 138.0 kV line loads to 229.2% of its 267 MVA rating. Project DFAX: 54.5%.

The SENECA / MCKEE RUN 138.0 kV line is overloaded to 215.7% of its 209 MVA emergency rating for the loss of the LIMON / BRISTOL 138.0 kV. The project contributes 34.9% DFAX (375.3 MW) to this overload.

The BRISTOL / LIMON 138.0 kV line is overloaded to 202.6% of its 258 MVA emergency rating for the loss of the SENECA / MCKEE RUN 138.0 kV. The project contributes 46.2% DFAX (496.9 MW) to this overload.

WARREN F SAM BEASLEY GENERATION STA / VAN SANT STA 138.0 kV L: 199.0% loading (212 MVA) following the outage of the ALTAVISTA / BRISTOL 138.0 kV L; DFAX = 45.5%.

For the loss of ALTAVISTA / BRISTOL 138.0 kV, the MAIN / GILROY FOOD 138.0 kV loads to 185.1% of its 195 MVA rating. Project DFAX: 27.5%.

The WARREN F SAM BEASLEY GENERATION STA / WHEATLAND STA 138.0 kV L is overloaded to 182.9% of its 223 MVA emergency rating for the loss of the ALTAVISTA / BRISTOL 138.0 kV. The project contributes 29.2% DFAX (314.6 MW) to this overload.

VAUGHN / MILFORD SOLAR FARM SUBSTATION 138.0 kV: 175.2% loading (219 MVA) following the outage of the ALTAVISTA / BRISTOL 138.0 kV; DFAX = 27.5%.

The MILFORD SOLAR FARM / MAIN SUBSTATION 138.0 kV is overloaded to 165.9% of its 230 MVA emergency rating for the loss of the BRISTOL / ALTAVISTA 138.0 kV. The project contributes 27.5% DFAX (295.3 MW) to this overload.

AVOCA / CHELSEA 138.0 kV: 165.7% loading (201 MVA) following the outage of the ALTAVISTA / BRISTOL 138.0 kV line; DFAX = 27.0%.

The VAUGHN / AVOCA 138.0 kV line is overloaded to 164.2% of its 247 MVA emergency rating for the loss of the BRISTOL / ALTAVISTA 138.0 kV. The project contributes 27.5% DFAX (295.4 MW) to this overload.

For the loss of LIMON / BRISTOL 138.0 kV Line, the SHOALS / RINGGOLD 138.0 kV L loads to 153.3% of its 186 MVA rating. Project DFAX: 25.5%.

SENECA / BANTA CARBONA 138.0 kV: 138.0% loading (277 MVA) following the outage of the LIMON / BRISTOL 138.0 kV; DFAX = 34.9%.

For the loss of RED LION ENERGY CTR / DELAWARE CITY PLANT 138.0 kV line, the KEENEY SUBSTATION / DELL CREEK 138.0 kV loads to 137.7% of its 196 MVA rating. Project DFAX: 5.4%.

For the loss of DELAWARE CITY PLANT / RED LION ENERGY CTR 138.0 kV L, the WILLOW PASS / CHURCHS FERRY 138.0 kV loads to 132.6% of its 196 MVA rating. Project DFAX: 14.0%.

The RINGGOLD / DOVER SUN PARK 138.0 kV is overloaded to 126.5% of its 213 MVA emergency rating for the loss of the LIMON / BRISTOL 138.0 kV. The project contributes 25.5% DFAX (274.6 MW) to this overload.

For the loss of N LEBANON / NINEMILE PT 138.0 kV Line 1, the NINEMILE PT / N LEBANON SUBSTATION 138.0 kV Circuit 2 loads to 123.3% of its 288 MVA rating. Project DFAX: 18.0%.

N LEBANON / NINEMILE PT 138.0 kV: 120.7% loading (294 MVA) following the outage of the NINEMILE PT / NORTH LEBANON 138.0 kV; DFAX = 18.0%.

The CHELSEA / SHOALS 138.0 kV is overloaded to 116.8% of its 196 MVA emergency rating for the loss of the ALTAVISTA / BRISTOL 138.0 kV line. The project contributes 18.6% DFAX (200.5 MW) to this overload.

For the loss of ALTAVISTA SUBSTATION / BRISTOL 138.0 kV, the WHEATLAND STATION / MCKEE RUN 138.0 kV loads to 112.8% of its 245 MVA rating. Project DFAX: 13.9%.

For the loss of DELAWARE CITY PLANT / RED LION ENERGY CENTER SUBSTATION 138.0 kV, the DELL CRK / CHURCHS FERRY 138.0 kV Line loads to 109.1% of its 142 MVA rating. Project DFAX: 10.0%.

For the loss of SHOALS / RINGGOLD 138.0 kV L, the MCKEE RUN / NRG ENERGY CENTER DOVER 138.0 kV loads to 106.9% of its 220 MVA rating. Project DFAX: 18.6%.

The N LEBANON / FAIRMONT 138.0 kV Line is overloaded to 102.8% of its 316 MVA emergency rating for the loss of the NORTH LEBANON / FAIRMONT 138.0 kV Circuit 1. The project contributes 17.8% DFAX (191.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ALTAVISTA / BRISTOL 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,789,000	100% (\$3,789,000)
ALTAVISTA / VAN SANT STATION 138.0 kV	Rebuild line with higher-capacity conductor	\$4,841,000	100% (\$4,841,000)
SHOALS / BRISTOL 138.0 kV	Rebuild line with higher-capacity conductor	\$7,214,000	100% (\$7,214,000)
AVOCA / VAN SANT STA 138.0 kV	Rebuild line with higher-capacity conductor	\$787,000	100% (\$787,000)
SENECA / MCKEE RUN 138.0 kV line	Rebuild line with higher-capacity conductor	\$8,710,000	98% (\$8,548,000)
BRISTOL / LIMON 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,715,000	100% (\$4,715,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WARREN F SAM BEASLEY GENERATION STA / VAN SANT STA 138.0 kV L	Rebuild line with higher-capacity conductor	\$15,925,000	100% (\$15,925,000)
MAIN / GILROY FOOD 138.0 kV	Rebuild line with higher-capacity conductor	\$5,988,000	100% (\$5,988,000)
WARREN F SAM BEASLEY GENERATION STA / WHEATLAND STATION 138.0 kV line	Rebuild line with higher-capacity conductor	\$11,202,000	100% (\$11,202,000)
VAUGHN / MILFORD SOLAR FARM SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$12,925,000	100% (\$12,925,000)
MAIN / MILFORD SOLAR FARM SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$7,700,000	100% (\$7,700,000)
CHELSEA / AVOCA 138.0 kV	Rebuild line with higher-capacity conductor	\$12,939,000	100% (\$12,939,000)
VAUGHN / AVOCA 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,368,000	100% (\$8,368,000)
SHOALS / RINGGOLD 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,109,000	100% (\$5,109,000)
SENECA / BANTA CARBONA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,692,000	98% (\$2,642,000)
KEENEY / DELL CRK 138.0 kV	Rebuild line with higher-capacity conductor	\$2,376,000	100% (\$2,376,000)
WILLOW PASS / CHURCHS FERRY 138.0 kV	Rebuild line with higher-capacity conductor	\$1,964,000	84% (\$1,652,000)
DOVER SUN PARK / RINGGOLD 138.0 kV	Rebuild line with higher-capacity conductor	\$2,531,000	100% (\$2,531,000)
NINEMILE PT / N LEBANON SUBSTATION 138.0 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,502,000	46% (\$685,000)
NORTH LEBANON / NINEMILE POINT 138.0 kV	Rebuild line with higher-capacity conductor	\$1,080,000	46% (\$492,000)
CHELSEA / SHOALS 138.0 kV	Rebuild line with higher-capacity conductor	\$21,942,000	100% (\$21,942,000)
WHEATLAND STATION / MCKEE RUN 138.0 kV	Rebuild line with higher-capacity conductor	\$2,746,000	96% (\$2,637,000)
DELL CRK / CHURCHS FERRY 138.0 kV Line	Reconductor line	\$706,000	86% (\$610,000)
MCKEE RUN / NRG ENERGY CTR DOVER 138.0 kV	Reconductor line	\$537,000	100% (\$537,000)
FAIRMONT / N LEBANON 138.0 kV	Reconductor line	\$2,065,000	46% (\$942,000)
	Total allocated to project		\$147,096,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.